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ESSAY
That Interactive Thing You Do 3 MESSAGE FROM NACIS'
Mic/we/ P. Peterso11 PRESIDENT
FEATURED ARTICLES It is a pleasure to serve as the 1997-
Beyond Graduated Circles: Varied Point Symbols for 6 98 president of NACIS.
No, really,
Representing Quantitative Data on Maps I am serious! (as Dave Barry would
Cy11thia A. Brewer and A11drew /. Campbell say). After well over a decade
of
association with the organization,
Animation-Based Map Design: The Visual Effects of 26
serving now and then as a member
Interpolation on the Appearance of Three-Dimensional Surfaces
of the board of directors or the CP
Steplre11 Lm1i11, So11ja Ross11111, a11d Slraw11 R. Slade
editorial board, I finally have the
pleasure of serving as NACIS' vice
Decision-Making with Conflicting Cartographic Information: 25 president and

cartographic perspectives
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Number 29, Winter 1998 cartographic perspectives 3
essay

That Interactive Thing You Do

Michael P Peterson
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The incorporation of interaction in the display of maps may be viewed as a major accomplishment of the computer-era in cartography. Certainly, interaction has pervaded all forms of mapping, whether it is with a database of a geographic information system, a multimedia atlas on a CD-RO:vt:, or street maps on the World Wide Web. A characteristic shared by all of these forms of mapping is the control that the user has over the resultant map.

However, interactivity is not new in cartography. In fact, it may be as old as cartography itself. While \.Ve don't know when the first map was made, it was very likely a product of an interaction between two individu als. It may have been much like the "paper-napkin map" of today-the kind of map that is drawn on any piece of scrap paper when words fail as one tries to explain where something is located. A common characteristic of these maps is that the person for whom the map is being made will ask

How can interactive maps be made better? A conversation is a good metaphor to use in understanding what interaction means with a map. An important aspect of a good conversation is that each participant responds in some way to what the other has said. A bad conversation is characterized by no response, a response that has nothing to do with the topic of the conversation, or does not relate to the previous point that was made.
An additional and important aspect of a conversation is that each participant is building up a database of what the other has said. The database can be fairly sophisticated at times. For example, I can remember stories that people have told me several years ago (although they have sometimes forgotten that they have already told me these stories -a flaw in their 'who-have-I-told-this-to' database). We have not reached this stage of sophistication in interactive cartography. The system rarely remembers what it has already shown. (The closest we get to this is the short-term caching structure of a World Wide Web browser.)
Viewed in the perspective of a conversation, the type of interaction that we have with maps on the computer is simplistic. It is somewhat like talking to a person for the first time over and over again. There is no database of the interaction. The computer doesn't remember anything. Because it doesn't remember, it cannot raise the sophistication of the

The NACIS Program Committee invites you to participate in this meeting by presenting a paper, poster or exhibit. Plan to organize a session, panel discussion, or workshop. A full range of computing facilities at the University of Wisconsin-Milwaukee will be available for workshops. We look. forward to featuring the current challenges confronting map librarians and cartographers.
Potential topics include:
handling digital map-library data
-7
compiling metadata
-7
managing the practical constraints of budgets
-7
-7 exploring data using visualization & virtual
XVIII reality
integrating diverse data sources for map
ANNUAL MEETING -7
production
marketing cartography services
-7

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Beyond Graduated Circles: Varied Point Symbols for Representing Quantitative Data on Maps

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Pe11nsylz1a11ia State University
U11iversity Park, PA 16802
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E11id, OK 73705

Graduated point symbols are viewed as an appropriate choice for many thematic maps of data associated with point locations. Areal quantitative data, reported by such enumeration units as countries, are frequently presented with choropleth maps but are also well suited to point symbol representations. Our objective is to provide an ordered set of examples of the many point-symbol forms used on maps by showing symbols with linear, areal, and volumetric scaling on repeated small maps of the same data set. Bivariate point symbols are also demonstrated with emphasis on the distinction between symbols appropriate for comparison (sepa rate symbols) and those appropriate for proportional relationships

examined. Point symbols are appropriate for quantitative data at point locations and for areal units. They are not affected by the physical size of enumeration units (large units produce large symbols on choropleth maps) and can be used to illustrate a spatial distribution while maintain ing recognizable geographic boundaries (unlike contiguous cartograms). Widely spaced point symbols in regions of large enumeration units (such as counties of the western U.S.) produce the appropriate sense of sparse data and they contrast well with dense clusters of symbols in areas of small units (such as eastern U.S. counties). Thus, point symbols preserve the spatial structure of the entire map without the problems associated vv·ith choropleth maps or cartograms.

The paper begins with a review of literature on graduated symbols to update the reader on this long research tradition in cartography and psychology. The remainder of the paper is devoted to a description of the v<iriety of symbol designs possible and the nuances in symbol designs that make them appropriate for different types of bi\'ariate map data. A wide \'ariety of symbols are illustrated with the hope that these examples •.vill inspire and encourage the authors of maps and GIS displavs to produce a wider ,·ariety of engaging and enlightening data displays well-suited to communicating and exploring their geographic information.

map space and, in the experimental context, promotes an increase in the variability of subject responses. (p. 35)

In analyzing intra-versus inter-subject variation in circle size estima tion, Griffin pointed out the ineffectiveness of apparent-value scaling for increasing the estimation accuracy of subjects who performed poorly. He also noted that apparent-value scaling impaired those subjects whose estimates were most accurate. In contrast to Griffin's results, Olson's (1975) earlier work with circle-size and dot-density estimation examined the interaction between apparent-value scaling and training in the form of practice and feedback about correct answers (for commentary see Will iams, 1977, and Olson, 1978). She concluded that the combination im proves both the accuracy and dispersion of subject estimates, unlike results for either scaling or training alone. Assuming that a map legend functions as training in a limited way by providing example sizes with associated data values, apparent-value scaling may produce improvement in graduated-circle estimation for complete map-reading contexts that is not seen in simpler experimental contexts.

Interactive and animated mapping facilitates examination of multiple symbol scalings. Slocum and Yoder (1996) described student use of Visual Basic to animate series of graduated-circle maps. One approach was to

tions of psychophysical research in cartography (Olson 1979; Fraczek 1984), and Gilmartin (1981a) discussed the inextricable link between cognitive and psychophysical study. Peterson (1985, 1987) examined mental images of maps and comparison of pattterns on point symbol maps, rather than comparison of symbols. MacEachren's How Maps Work (1995) is a current, in-depth analysis of human cognitive processes used to understand map symbols and map patterns.

Beyond Circle Size Estimation

Groop and Cole (1978) studied the relative effectiveness of cut-out versus overlapping graduated circles, since many graduated circle maps include dense clusters of circles. Their analysis showed that estimation errors associated with cut-out circles were significantly greater than those associated \vi th overlapping circles. Cut-out circles are seldom used today because of the ease of constructing overlapping circles with the computer. However, Dent (1996) noted that they "add a three-dimensional (plastic) quality to the map" (p. 177). Groop and Cole also questioned the effects of clustering on the accuracy of size estimation. Gilmartin (198lb) found that circle size estimation was affected by neighboring circles and that the effect can be minimized by including intervening linework such as bound aries separating enumeration units. Dent noted the current lack of a

representation of comparison and proportional relationships within multivariate graduated symbols.
In 1926, Eells examined "The Relative Merits of Circles and Bars for Representing Component Parts." He noted commonly held criticisms of using circles to illustrate component parts (what we refer to as propor tional relationships) including: difficulty in rapid and accurate estimation of values, inaccuracy of estimation due to their areal nature, and the suggestion that "while their popularity and psychological appeal are grudgingly admitted their use is 'an insult to man's intelligence'" (p. 122). However, Eells' study concluded that circle diagrams divided into compo "while their popularity and nent parts were easily and accurately read, that accuracy increased with
the number of subdivisions of circles but decreased for subdivided bars, psychological nppeal are
and that the use of circle diagrams was "worthy of encouragement" (p. grudgingly ndmitted their use is
132). Croxton and Stryker's 1927 follow-up study supported Eells's
'rm insult to mnn 's findings and suggested that for illustrating 25, 50, and 75 percent relation
intelligence'" ships, circles worked significantly better than bars. Croxton and

and Gilmartin examine symbols representing sets of four variables with attention to the multiple purposes of multivariate map symbols: from 'what' and 'how much' at the local level (which may be better represented with separable dimensions) to regional patterns and correlations between variables (which may be better represented with integral dimensions).
As an example of multivariate symbol research, Slocum (1981) exam ined two-sectored pie graphs. Specifically, he measured the just-noticeable difference for sector size and accuracy of sector size estimation. He found that subjects could not discriminate between sector size differences of less than nine degrees, and that sector sizes were estimated within a three percent margin of error. He suggested rounding data to the nearest five percent, rather than one percent, before drawing sectors. His study was, in part, a response to Balogun (1978), who suggested the use of decagraphs (ten-sided polygons) for representing proportions. Examples of decagraph use can be found in the Atlas c~f Ncufo1111dla11d a11d La/1rador, ·where they are
used to represent economic data (McManus et al. 1991, plates 10 and 19).
Point Symbol Discussion in Cartographic Textbooks
Cnrtosrnphy: Tlre111atic Map Design by Dent (1996) and Elements of Cartog raphy by Robinson et al. (1995) are two widely used collegiate texts

classes, for example, may be a useful compromise between the comfort of easy identification and an accurate overview of relative magnitudes across a map. Similarly, low symbol dimensionality improves symbol-size
"A hybrid approach with estimation but may hinder map pattern interpretation. The larger range in
symbols scaled to means of data sizes of linearly-scaled symbols (bars, for example) causes symbols to
extend farther across the map than higher-dimension and more compact classes, for example, 111ay be a
symbol forms (areal and volumetric symbols such as squares or cubes) useful compromise ... "
representing the same data. Adjustments within a map series, such as equalizing symbol areas among maps before comparison (Peterson 1985) or using constant symbol sizes for minimum and maximum data values (Monmonier 1977), assist pattern comparison but interfere with compari son of data magnitudes between maps. Another challenge is choice of multivariate mappings with separable symbol dimensions (which may encourage comparisons within symbols and within variables) versus integral dimensions (which may improve understanding of patterns in correlations between variables). Although researching the endless variety

Number 29, Winter 1998 13
a. Areal Table b. Choropleth
1993
VENEZUELA
Disbursements
2137
D
GUYANA 69
69 Disbursements •
mapped are the
COLOMBIA
amount of each
1697 country's 1993 debt 662
•
flow that is drawn on
loan commitments.
ECUADOR Reported in millions 1551 -2137
662 I BRAZIL of U.S. dollars •
12195
~ Source: 12195
PERU World Bank 1994

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- 0~ E
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5.0
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'' 1.6
c. Graduated d. Graduated
Squares Ellipses
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..;_ a. Graduated
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100 500 5000
b. Graduated
Spikes
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I

units having generally elongated shapes. These
a. Graduated
ellipses are also included for comparison to a later
Spheres
bivariate use (Figure 6g). Magnitudes represented
with a square grid (Figure 2e) may aid accurate
100
interpretation because individual cells can be
counted.
500 Another areal symbol, the 'money bag' picto
gram (Figure 2f) offers an example shape that
more literally reflects the data, captures the
reader's attention, and may be scaled to represent
relative quantities. Pictograms are particularly
well suited when 'catching' the reader's eye is a
priority, making them common in magazines,
newspapers, and educational material. The New
b. Graduated Stnte of the World Atlns (Kidron and Segal 1984)
Cubes offers good examples of the unlimited possibilities
of pictogram use. This atlas and later editions (for

1961; Brewer 1982; Dubery and Willats 1983; Figure 5). Block Orthographic
piles (Figure -le) have long been associated with Raisz's talent
for cartographic symbolization (1939, 19-!8). D 1cm
cm
Co111pnriso11 Dntn: Disb11rse111e11ts n11d Pri11cipal Repnyments
1
The possibilities for comparing data with point symbols on
bi\'CHiate maps are limitless. The focus will be on several
Oblique examples
combinations of circles and squares, as well as bars, ellipses, and
cavalier cabinet
cubes. The large number of arrangement options available using 4/11 1
common graduated circles and squares (side by side or ·H=71
m·crlayed) offer different visual impressions while illustrating
LlJ.
the same data (Figure 6a, b, d, and f). Overlapping the symbols LJ)45
produces a more integral symbol that involves pattern recogni usual usual
tion because differences in relative values produce different
overall symbol forms (compare the mostly black symbol for

---
a. Adjacent b. Graduated
Graduated Square
Squares Overlay
Pnncipal
Pnncpal Dlsbursemenls Pavments
D sbursements Payments
111100 • 100.
11
soo
11 500
c. Adjacent d. Graduated
Graduated Circle Overlay
Semicircles
PnncipaJ
Payments
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a. Nested b. Graduated
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~ -
F1.x1m• 7 \\,lumetnc bivariate point .,ymbols for both comparison (il) and proportional (b) rl'iallonships, The same dat,1 represented in Figure'> 6 and 8 are u.,ed for the cube~ in 7a and b rc~pcctively,
Proportional Data: Public and Private Components of Disbursements
Mapping proportional data differs substantially from mapping com parison data because the smaller of the two data sets is always part of the larger set, unlike the disbursement and repayment data used in the comparison examples (Figure 6 and 7a). In Figures 7b and Sa, c, e, and g, graduated symbols are scaled to represent total disbursements and are segmented by the proportions of loans to public and private entities. Symbols of constant size in Figures Sb, d, and fare segmented to represent only the proportions of the disbursements' compositions, not the total

a. Graduated ,,-./?S· b. Segmented
Segmented ·.. ~___...,__~ Bars
I Bars I / /) ~
Percent
of 1993 Loan
I Disbursements
to Private
Debtors
~ % Private
Percent of
% Public ~ ~ 1993 Loan
• Disburse
f ments to
I' Public
100 500 5000 Entities
'''''
". ~
c. Graduated d. Segmented
·- Segmented Squares
Squares \

points in their symbol designs, which assists accurate percentage esti mates. The symbol form in Figure 8b is known to statisticians as a framed rectangle (Cleveland 1985, p. 208; Dunn 1987, 1988), and Monmonier (1993, p. 65) describes its usefulness in representing both absolute magni tudes and intensities. Dunn (1987) also suggests scaling the widths of the rectangles to total data values (to total disbursements for Figure 8b). In a conversation about Figure 8, MacEachren mentioned a successful student project from an ACSM design competition that divided graduated squares into ten-by-ten grids such that each cell represented one percent, and each row ten percent, of the whole for all symbol sizes (rather than using grid cells of a fixed size; a hybrid of Figure 8c and 2e).

The work of students new to cartography has been an inspiration. With

CONCLUSION

modern computing at their disposal and freedom from the conventions of traditional thematic mapping, students are producing a wide variety of creative symbols for mapping data usually symbolized by graduated circles or squares. For example, a recent student assignment with eco nomic data produced creative ideas ranging from a proportionally scaled \ilonopoly game character (with pockets turned out or clutching money bags\ to weight scales drawn tipped to illustrate balances between dollar

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Animation-Based Map Design: The Visual Effects of Interpolation on the Appearance of Three-Dimensional Surfaces

Stephen Lavin

Computer animation is a potential aid in map design, because it pro

Sonja Rossum

vides a means for quickly reviewing many design alternatives. This

research is a conceptual exploration of one aspect of animation-based

Shawn R. Slade

design: The effect of inverse-distance weighting on the visualization of

Department of Geography

three-dimensional maps. The primary variable examined is the

University of Nebraska

inversedistance weighting exponent. Changing the exponent in small

Lincoln, NE 68588-0135

intervals allows the creation of a series of three-dimensional maps that

can be assembled, and played back as a frame animation. In this proce

dure, cartographers can view the visual effects of various exponents on

the resulting surfaces. Design-based animations allow mapmakers to

2ao0
Vertical Angle. An angle of 90 degrees Vertical Exaggeration. A value of zero would Horizontal Rotation. A rotation angle of 360
places the viewer directly above the surface show a totally flat surface. degreesplaces the viewer due east of the surface.
F1~1111' I. Ori<'llf11/io11 rnrin/>les i11 tlnl'c-dm11·11~i<mnl mnpp111s example, it is easier to envision what \·viii happen to a surface if the angle of \'iew above the horizon is changed from 30 to 50 degrees than it is to predict the ,·isual effects of changing the inverse-distance weighting exponent from one to three. Consequently, this research is concerned with the more difficult case of visualizing the effects of interpolation decisions.
"A11i111atio11s are potentially
While it is true that contemporary three-dimensional surface-producing valuable aids in selecting opti
software allows experimentation with the effects of design choices on map mum wnys of designing three
appearance, this is usually accomplished one map at a time and normally requires many iterations before a satisfactory map is created. For example, di111e11sio11al maps, representing
SURFER (Golden Software 1990), a popular isoline and three-dimensional

focus is just one aspect of three-dimensional mapping, it is not unreason able to assume that animation can be used to explore variations in unit value in dot mapping, or the visual effect of varying isoline intervals, for example.

Visualization

Geographic visualization is an act of cognition through which mental images of geographic data which have no visual form are developed (Peterson 1995, MacEachren 1995). This notion of visualization can be extended to the realm of computer cartography in which data stored in computer memory can be displayed visually. Cartographic visualization "Clearly, ani111atio11 e11ables of geographic data can facilitate examination and explanation of processes
mapmakers a11d map-users alike, which evolve over time, over space, or a combination of both (Campbell
and Egbert 1990). It is also a concept which focuses on the exploration of to perceive differe11ces which
data rather than data presentation. As such, cartographic visualizations may otherwise go undetected
created with exploratory data analysis in mind are generated as animation
who11 viewi11g a series ofs tatic sequences used to identify trends and

The challenge of interpolation is to choose a technique which creates a plausible model of the variation between control points. Selection of the most appropriate method depends on the type of data used, the amount of computational time available, and the degree of accuracy required (Lam 1983), although this last factor is difficult to anticipate in advance. A wide variety of algorithms have been developed for point interpolation. For example, surfaces constructed by kriging, minimum curvature, and inverse-distance weighting interpolations are shown in Figure 2. As can be seen, variations in interpolation method can produce pronounced visual differences. Kriging is a regional variable theory technique that assumes an underlying linear variogram, and because of this it can pro duce an estimate of error associated with an interpolation (Burrough 1986, Sampson 1975). Minimum curvature keeps the grid surface fixed at known control points while iteratively applying an algorithm to smooth the grid ded surface. Its primary benefit is that the resulting surface is constrained to pass through all control points (Golden Software 1990). Other interpola tion methods include splines, polynomials, Fourier Series, Power Series Trend, and distance-weighted least-squares (Lam 1983). Each technique hils a different way of modeling the variation between control points and produces a map surface with a distinctive appearance. Although it is

I (Zi/d/vt) where:
Z = z value
I: (l/d.wt) d =distance
wt = weighting value
I
Equation 1. /11verse Distance Weighting Eq11atio11
on the local average also increases while the influence of more distant points decreases. Emphasizing nearby points creates more peaks and valleys in the surface in contrast to the smoother appearance resulting from the use of a low weighting exponent which gives distant points more numerical equity in the computation.
A single set of data values located at control points can be subjected to inverse-distance weighting interpolation and then, by incrementally in creasing (or decreasing) the weighting exponent in small numerical steps, a sequence of maps showing weighting exponent effects on visual surfaces can be generated. Maps thus generated can be played back as a frame-by
"Maps thus generated can be
frame animation, and the effects of differences in the weighting exponent played back as a frame-by-frame
can be observed. This can be done for either isoline or three-dimensional

WT=1 WT=1
WT=2 WT=2
WT=3 WT=3
WT=4 WT=4
WT=5 WT=5
WT=6 WT=6
METHOD=ALL METHOD=NORMAL
Figure 4. Frames scle«tcd from a11i11111lio11;;. A11i111atio11s /inpr a Iola/ vf120 fr11111e:<. The method
,.de,.red lo as ALL 11'.'cs ali co11trol poi11I «alue,; i11 iim:rs,• disfm1cc <i>;ig1ttii1g. The mctlrvd rcfa,wl to
"One long-standing, useful,
as NOR.lvlAL 11:<es a rcgii•11al selecf1011Lfco11trol17vi11ts based 011 a sclecf1,111of11ear-11eighl>ors. \.\Tis
f/1e weigl1liJ1S <'Xi'OJICl/f. and often violated convention in
graphics and cartography
distributions of control points because equal weights are assigned to each maintains that the selection of
point even if it is in a cluster (Lam 1983). In reality, the mapmaker does not
symbols for maps and graphs

Robinson and Petchenik 1976, Dobson 1973). When studying the effects of inverse-distance weighting in three-dimensional animations, continuous data is most appropriate for visualization. Precipitation is a truly continu ous geographic phenomena, and for that reason was chosen for anima tions in this study.

Individual animation frames (maps) were constructed in SURFER by entering control point x, y locations, and corresponding z-values (precipi tation) for the seventy climate stations, and then computing the inverse distance grid interpolation. For each successive frame, the weighting exponent was increased by a value of 0.05. This fairly small increment was chosen to produce a smoother animation, and one which lasted a sufficient length of time to allow examination of the visual effect of increasing weights. An attempt to utilize an increment of 0.1 was made but rejected because the animation appeared too discontinuous and jumpy between successive frames. Weighting exponents ranged from 0 .05 to 6, stopping at 6 because higher weights produced little change in the three-dimen sional surface. This exponent range and numerical increment yielded a total of 120 frames.

Most visual parameters were set and remained constant throughout production of individual frames for each animation. In order to examine

mately be determined by the author's purpose and goal, statistical consid erations, the intended audience, and the author's knowledge of the geographic phenomena being mapped. In this regard, design-oriented animations can assist authors in making more informed map selections. Animation-assisted map design is a promising tool for cartography, and

CONCLUSION

can be extended \Nell beyond animation of the visual effect of inverse distance weighting on three-dimensional mapping. Virtually all variables encountered in designing and constructing three-dimensional surfaces can be animated, including data search procedures used in gridding, surface rotation, vertical scaling, and horizontal angle of view. The greatest potential for this method lies in examining design decisions whose conse quences are difficult to visualize mentally, such as those dealing with the mathematics of grid interpolation. Jn a further example of related uses of

X=20

animation, Figure 5 shows frames from an animation in which only the density of the interpolated grid is varied. Mapmakers can examine this animation to determine an appropriate level of grid-based generalization for the final product. The first frame (top) shm.vs a surface that is made from a grid that h.loo coarse to give an adequate impression of a contin-

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Decision-Making with Conflicting Cartographic Information: The Case of Groundwater Vulnerability Maps

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Conflicting cartographic information can cause problems when used to support planning decisions. Creation of conflicting information is becoming more common as geographic visualization and modeling software are used to develop multiple maps that represent different views of the same data. This paper presents groundwater vulnerability mapping as an example of information conflicts of this type. Three different vulnerability models applied to the same test data produced radically different results. This information was presented to a group of local planners to examine how they would deal with the conflicts. Through this exercise it became apparent that each planner used highly

visual information should provide strategies for improving both carto graphic displays and users' interpretations of these displays. To examine this question, a map use exercise based on maps from different groundwa ter vulnerability models was presented to a group of local planners, elected officials, and groundwater specialists.

The first part of this paper describes the vulnerability models, the context for their use, and a more formal problem statement. An overview of the map use exercise and the results are then presented. Finally, issues pertaining to the strategies that decision makers employ for dealing with visual information conflicts are discussed. In approaching this study, it is hoped that concrete recommendations can be provided on how to resolve information conflicts between maps. However, the participants' disparate responses indicated that the fundamental role of maps in decision-making was at issue and that diverse approaches to problems of conflicting information were in operation. Parallels with Jung's psychological typol ogy are drawn upon to help understand variations in decision-making strategies. Finally, avenues for further research are suggested.

Conflicting Visual Information in Cartographic Representation and Planning

The potential for conflicting cartographic representations appears in

Planning involves the design and consideration of a series of alterna tives. Monmonier (1991) reviewed the use of maps in zoning and environ mental protection. Ylaps serve as tools for making planning decisions and communicating a plan's spatial implications. In DiBiase's (1990) con tinuum of visual thinking and visual communication, planning maps operate in both the private and public realms. The maps are used in planning policy decisions, essentially a thinking task, and in presenting planning policy decisions, essentially a communication task. In this context, the power of maps to persuade should not be overlooked (Harley 1989, Wood 1992). The 'official' nature of maps often operates at a subcon scious level.

Monmonier (1991, 71) notes that "without maps, planning would be chaotic, furthermore even with maps, many would argue that planning is chaotic." In addition, he comments that errors in map compilation can be significant and that the same information is often used to develop differ ent plans (Monmonier 1991). Pla1ming scenarios often involve creating different conceptual models about a future reality. Different scenarios can often produce markedly different results. The 'stability' of results may be re-framed with the concept of 'model stability.' Model stability describes how representations change through the use of different models for

hydrogeologic characteristics and contaminant fate, such as dilution, breakdovvn, absorption, and volatilization (Geraghty & Miller, Inc. and ICF, Inc. 1990, 6-1) .

.\1ost of the models that have been applied in the context of GIS have been hydrogeologic setting or parameter weighting/ scoring models. DRASTIC (Aller et al. 1987) is a general-purpose, parameter weighting model that was developed for the US EPA for vulnerability assessments at county and state levels. Several other models have been developed for state-wide assessments, including the Wisconsin Susceptibility Model (WISM) (Schmidt 1987), which uses weighted hydrogeologic characteris tics to model vulnerability, and an aquifer vulnerability model of Michigan (Lusch et al. 1992), which uses hydrogeologic characteristics for determin ing relative risk. All of these models use a combination of data layers that include soil, subsurface geomorphic characteristics, geology, recharge rates, and topography.

Little research has addressed differences in model performance. The authors of the models often state that results from different models are not directly comparable (Aller et. al 1985; Schmidt 1987). Furthermore, Mer chant (1994) notes that there is little validation of these models in the context of GIS and that there are a number of questions concerning the

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Correlation Coefficients Coefficient of Areal Correspondence
(standardized unclassed scores) (classed scores)
SCAM-DRASTIC 0.049 None 22.4%
DRASTIC-WISM 0.279 Partial 70.3%
WISM-SCAM 0.311 Complete 7.3%
Ta/1/c 1. Rclatio11slrips /1efwee11 Gro1md1mfer V11l11erability Models
participants had significant experience with groundwater vulnerability models and mapping.
The exercise had the participants identify potential sites for houses using the information on groundwater vulnerability for each of three
"The exercise had the partici
different maps. Each map was divided into four quadrants. Participants
pants identify potential sites for were asked to assume that all sites were equally accessible within each
/rouses using the information on quadrant and that each site would have its own well and septic system.
grou 11dwater vulnerability for For each map, they also rated the vulnerability for each quadrant, chose
the best location for a house in each quadrant, and selected the overall best
each of three different maps."

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Groundwater Vulnerability Area 3
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- Moderate High
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"'SIGNIFICANTLY DIFFERENT at p < .05
Table 2. V11/11aal1i/1t,11 Rati11gs /1ct11>ee11 Map Pairs by Q11adrn11t (ANOVAi
context, the information content was reasonably stable. In other words, the maps produced the same result for the determination of the best quadrant for a house although the maps were different.
The best location for a house within each quadrant varied by map (Figure 3.). It is not surprising that in areas where locational choices are constrained, more clustered patterns occur. SCAM and WISM models provided high constraints, and therefore the participants' responses tended to cluster. These clusters occurred in different locations in all

The importance of the map and spatial approaches to the problem DRASTIC varied greatly. These approaches could be characterized by a continuum from aspatial to spatial (Figure 4.); with less reliance on the map defining the aspatial end and more reliance on the map defining the spatial end of the continuum. The implicit assumption at the aspatial end of the con tinuum is that the map is unnecessary. Common sense, unreflective opinions, or rules-of-thumb fall at the aspatial end of the continuum, because they often do not explicitly consider local ecological conditions. For example, "everyone knows that you don't put your well next to your septic system." While initially these may appear to be spatial approaches, these approaches do not involve the evaluation of specific site characteris tics. In certain circumstances, for example, in areas with deep wells, the 'rule' may have little practical meaning. The issue is that one does not need map-based information to apply the rule. The implicit assumption at O Hou•e locwon (DRASTIC)
the spatial end of the continuum is that a map is needed, but that the existing models need to be refined.
SCAM
The question arises, why do people view the utility of maps (and models) with such divergence? There are many different possible explana

Aspatial Spatial for degrees of closure. Judging types prefer closure using the available
data, and perceiving types defer closure desiring more data.
Returning to the idea of how participants made decisions using the
maps, it is possible to arrange several of the psychological functions along
a continuum from aspatial to spatial (Figure 5.). This provides a potential
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for explaining the variations in approaches to conflicting
::pnf111/ dl'Ci,;1,111-111nki11s. information on maps. Common sense
approaches align with intuition and
feeling and generally appear to be aspatial, whereas modeling approaches
align with sensation and thinking and appear to be spatial. Spatial and
aspatial are defined solely on the basis of whether or not a map is neces
sary to the decision-making process. Furthermore, intuitive types prefer
approaches that are figurative and employ approximations; in contrast,
sensing types prefer approaches that are literal and employ detailed facts
(Kroeger 1992, 33). Common sense involves "unreflective opinions" (G. &
"Spatial and aspatial are defined C. Merriam 1987, 266), rather than careful
evaluation of facts. As noted
earlier, since common sense does not rely explicitly on site-specific criteria

information as a cartographic problem; the problem is not necessarily a question of 'how maps work', but instead, 'how people work with maps'.
In this paper, results were presented from an experiment that investigated
SUYfMARY AND
how decision-makers respond to conflicting cartographic information CONC
IU SIOl\'S
using a series of conflicting groundwater vulnerability maps for one area. Thirteen local decision-makers participated in the experiment, and input \·Vas collected from them as to how they would deal with information conflicts in a planning situation. The decision-makers then discussed the maps and information conflicts in a group setting to examine how group dynamics might influence the decision-making process. From this, a continuum of strategies was developed that ranged from the aspatial approaches, e.g., discard the maps and use common sense, to spatial approaches, e.g., build a new spatial model that "works."
When the research was initiated, it was viewed as a problem of carto graphic visual information processing and it was anticipated that the participants would discuss methods for representing conflicting informa tion. Hmvever, the issue turned very quickly away from the original maps
"Howeyer, the issue turned ztory

issues pertaining to psychological typology. All omissions and interpreta tions however remain our responsibility.

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numerous health conditions and of equal size. These graphs show
reviews
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Conclusion: The Louisiana State University search focus by concentrating on
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1950s and 1960s and the current In general, access to collection In addition to the daily map li
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(FDLP). The current FDPL material while other local finding aides are long term rehousing, retrospective
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The University of Iowa's Map Col from another library on campus, As you can see, the path to get
lection is the largest map collection and were able to purchase very ad ting the three machines was not
in Iowa, with more than 176,000 equate hardware to run Arc View 3. clear-cut, the results came to us in
sheet maps, as well as: copies of At that point, ArcV iew was relo phases, but the focus audience was
Sanborn fire insurance maps for cated from documents to the Map always the same. The administra
more than 1,000 Iowa cities from Collection to run on the new ma tors at the University Libraries are
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county atlases for the period from two more desktop mapping work Collection. We had earlier made
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requests to university offices

Secretary:JamesO. \!linton Jeremy Crampton (10 /99) NACIS Board Meeting
Map Library Department of Geography January 10, 1998
University of Tennessee George Mason University Lexington, KY
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Knoxville, TN 37996-4066 (703)993-1217 Present: Andrews, Baruth, Brewer,
(423)97-l-4315 fax: (..+23)974-0560 email: jcrampto@gmu.edu Chu,
Crampton, Gilmartin, Krejcie,
email: jim-minton@utk.edu Meacham, Minton, Nelson,
Term expires: October 99 Valerie Krejcie (10 I 98) Patterson, Peschel,
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Treasurer: Sona Andrews Skokie, IL 60076
Univ. of Wisconsin-Milwaukee (847)677-1881 Michael Peterson recognized
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P.O. Box 413 email: t-krejcie@nwu.edu Gilmartin as the newly-elected
Milwaukee, WI 53201 President of ACIS and turned
(-ll-1)229-6057 fax: (414)229-2-181 James E. Meacham (10/99) over the
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gested for CP Editorial Board. Spring Board Meeting tinuity and e'\perience provided by
Buttenfield offered to send flyers to Peterson noted that the experience Chris, Sona, and Susan form a
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§ to promote communication, coordination, and cooperation among the producers, disseminators, curators, and users of cartographic information;

§ to support and coordinate activities with other professional organizations and institutions involved with cartographic information;

§ to improve the use of cartographic materials through education and to promote graphicacy;

§ to promote and coordinate the acquisition, preservation, and automated retrieval of all types of cartographic material;

§ to influence government policy on cartographic information.

NACIS is a professional society open to specialists from private, academic, and government organizations throughout North America. The society provides an opportunity for Map Makers, Map Keepers, Map Users, Map Educators, and Map Distributors to exchange ideas, coordinate activities, and improve map materials and map use. Cartographic Perspectives, the